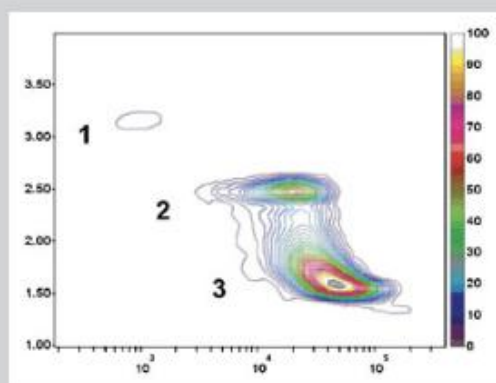


Summary: The chromatographic analysis of hydrophilic copolymers is complicated due to the fact that in most cases aqueous eluents must be used. In aqueous eluents different polar and ionic effects may disturb the selective interactions between the macromolecules and the stationary phase making it impossible to separate such copolymers with regard to chemical composition. Therefore, 2D chromatography combining a separation according to composition with a separation according to molar mass has been applied mostly to polymers that are soluble in organic solvents. The present contribution describes experimental approaches to analyze such hydrophilic copolymers by 2D-chromatography. For a model polymer system resulting from the copolymerization of methacrylic acid and a poly(ethylene glycol) macromonomer, it is shown that different analytical techniques including SEC, LC-CC, MALDI-TOF MS and 2D chromatography can be used to analyze the different parameters of molecular heterogeneity of such copolymers.



2D separation of poly(MPEG-MM 2), 1st dimension: LC-CC, 2nd dimension: SEC.

Multidimensional Chromatographic and Hyphenated Techniques for Hydrophilic Copolymers, 1

Analysis of Comb-like Copolymers of Ethylene Oxide and Methacrylic Acid

Martina Adler,¹ Frank Rittig,² Stefan Becker,² Harald Pasch^{*3}

¹PSS Polymer Standards Service GmbH, In der Dalheimer Wiese 5, 55120 Mainz, Germany

²BASF Aktiengesellschaft, Polymer Research, 67056 Ludwigshafen, Germany

³Deutsches Kunststoff-Institut (German Institute for Polymers), Schloßgartenstr. 6, 64289 Darmstadt, Germany

E-mail: hpasch@dkl.tu-darmstadt.de

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Introduction

Polymers are highly complex multicomponent materials. They are composed of macromolecules varying in chain length, chemical composition, and architecture. Depending on the composition of the monomer feed and the polymerization procedure, different types of heterogeneities may become important.

One very efficient approach for the analysis of the molecular heterogeneity of complex polymers is their chromatographic separation by combining different separation mechanisms. A typical experimental protocol includes the separation of the sample according to composition yielding fractions which are chemically homogeneous. These fractions are transferred to a size-selective separation method and analyzed with respect to molar mass.

As a result of this two-dimensional (2D) separation, information on both types of molecular heterogeneity is obtained. Another useful approach is the combination of a selective chromatographic technique with a powerful spectroscopic method like NMR or MALDI-TOF mass spectrometry.^[11]

There are numerous selective modes for LC of polymers, including liquid chromatography at the critical point of adsorption (LC-CC)^[2–4] and isocratic or gradient HPLC.^[5,6] Using these techniques polymers can be separated selectively with regard to chemical composition or functionality, as has been shown for macromonomers, random and block copolymers, and polymer blends.^[6–9] LC-CC and gradient HPLC are promising first dimensions in a 2D chromatography setup. These can be combined with size exclusion chromatography (SEC) in the second dimension that yields the corresponding molar mass distribution.

So far, 2D chromatography has been applied mostly to polymers that are soluble in organic solvents. Very typically, in the first dimension binary eluents are used, e.g., tetrahydrofuran-hexane, while in the second dimension THF is the eluent. At present we are not aware of any applications, where aqueous mobile phases are used in both dimensions when LC-CC and SEC are coupled in the 2D experimental setup.

There are several challenges related to the use of aqueous mobile phases in polymer chromatography. The types of polymers that are to be analyzed using such conditions are water-soluble polar or ionic copolymers. Even the SEC analysis of copolymers analyzed in aqueous mobile phases is not straightforward. In addition to the fact that only a very limited number of stationary phases is available for aqueous SEC, the experiments in most cases cannot be conducted in pure water. Due to the high polarity or the ionic nature of the polymers, electrolytes or ion-pairing reagents have to be added to the eluent to screen ionic interactions with the stationary phase, for more details see Mori and Barth.^[10] Salt solutions, on the other hand, can cause severe problems for the SEC detector.

As for HPLC in aqueous mobile phases, there are only a few applications in the literature so far.^[11–16] Regarding the analysis of water-soluble copolymers by HPLC methods and the determination of the chemical composition distribution of such copolymers the only publications relate to copolymers of ethylene oxide and propylene oxide,^[17] partially hydrolyzed poly(vinyl acetate)^[18] poly(styrene sulfonate) and poly(acrylic acid).^[19]

Recently, we reported on the analysis of copolymers that were prepared by grafting of methacrylic acid onto poly(ethylene glycol) by multidimensional chromatography.^[20] The graft copolymers were analyzed with regard to the complex chemical composition-molar mass heterogeneity by coupled HPLC-FTIR and 2D chromatography. We showed that the direct grafting of methacrylic acid onto

PEO results in reaction products that still contain large amounts of non-grafted PEO. In the present contribution we report on the analysis of comb-like copolymers of ethylene oxide and methacrylic acid, which were prepared through copolymerization of a macromonomer and methacrylic acid by interaction chromatography, MALDI-TOF mass spectrometry and 2D chromatography.

Experimental Part

Samples

The macromonomers MPEG-MM 1 and MPEG-MM 2 were commercial products of Sigma-Aldrich Chemie GmbH (Munich, Germany). The preparation of the comb-shaped homopolymer (sample 1) was accomplished as follows: A two-liter glass reactor equipped with a thermometer, a mechanical stirrer, a nitrogen purge device, a reflux condenser, a monomer addition pump, an initiator addition pump, and a chain-transfer agent addition pump was used. The monomer pump was charged with 668.00 g of macromonomer MPEG-MM 2 (50 wt.-% aqueous solution of methoxy poly(ethylene glycol) monomethacrylate with a number-average molar mass of 1000 g/mol), the initiator pump with a solution of 5.00 g sodium peroxodisulfate in 58.56 g water, and the chain-transfer agent pump with 29.22 g of a 40 wt.-% aqueous sodium bisulfite solution. The reactor was charged with 235.19 g water. With stirring, the internal air of the reactor was then replaced with nitrogen, and the reactor was heated to 100 °C under the nitrogen atmosphere. Simultaneously starting, the monomer was continuously pumped into the reactor over periods of 5 h, the initiator and the chain-transfer agent over a period of 5 h and 30 min, respectively. The internal temperature of the reactor was maintained at 100 °C for one more hour to complete the polymerization reaction. Finally, the resultant reaction mixture was cooled to RT and a pH 7 was adjusted by adding a 50 wt.-% aqueous sodium hydroxide solution.

The comb-shaped copolymers (samples 2–4) have been synthesized by copolymerizing methacrylic acid (MAA) with the macromonomer MPEG-MM 2 according to the same procedure as the homopolymer and were finally neutralized with 50 wt.-% sodium hydroxide solution. The following amounts of reactants were used, see Table 1.

After the individual determination of the solid content of each sample (all between 30 and 35 wt.-%) the samples were diluted with the eluent to a concentration of 1 g/L solids for one dimensional measurements and to 10 g/L solids for coupled methods.

Chromatographic System

A Shimadzu LC-10AD-VP HPLC system comprising a pump, an autosampler and a RI detector was used. For 2D experiments an evaporative light scattering detector ELS 1000 (Polymer Laboratories, UK) and an additional pump were added. The transfer of the fractions was carried out with a 8-port-2-position switching valve type ET8GW (Valco Instruments Co. Inc.) with two 100 µL sample loops. For data collection and

Table 1. Experimental conditions for copolymerization reactions.

Sample*	Reactor	Charge into monomer pump		Charge into initiator pump		Charge into chain-transfer agent pump
	Water	MPEG-MM 2	Methacrylic acid	Sodium peroxodisulfate	Water	40% Aqueous sodium bisulfite solution
	g	g	g	g	g	g
2 (90:10)	268.53	601.20	33.40	5.00	58.56	29.22
3 (80:20)	301.93	534.40	66.80	5.00	58.56	29.22
4 (70:30)	335.33	467.60	100.20	5.00	58.56	29.22

*(weight ratio MPEG-MM 2 to methacrylic acid).

processing the software package "WinGPC-Software" (Polymer Standards Service GmbH, Mainz, Germany) was used. Molar mass calibration was based on PEG. The injection volumes were 50 μ L for the one dimensional experiments and 100 μ L for the 2D experiments.

Columns

The LC-CC experiments were performed on a Knauer Nucleosil RP18-100 (250 $\times$ 4.6 mm inner diameter (i.d.)) HPLC column and the SEC experiments on a PSS Suprema linear M (300 $\times$ 8 mm i.d.) column. For the 2D measurements a Macherey-Nagel EC 250/4.6 Nucleosil 100-5 C18 (250 $\times$ 4.6 mm i.d.) and a PSS Suprema HighSpeed (50 $\times$ 20 mm i.d.) column were used.

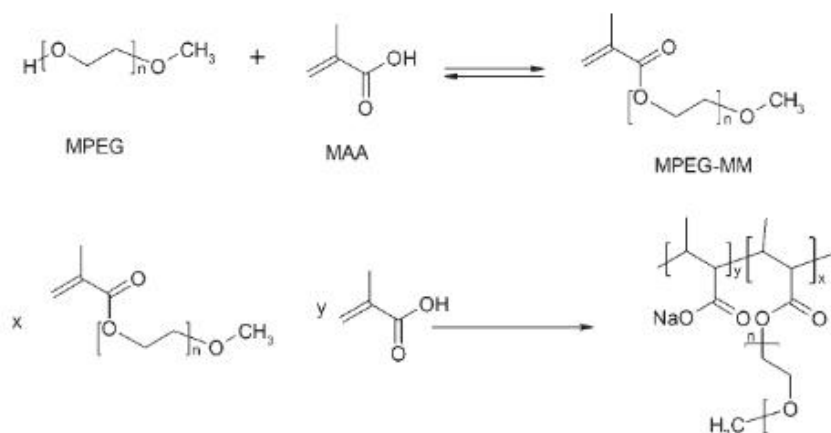
Mobile Phase

Methanol HPLC grade (Acros, Belgium); water in ultrapure quality (Simplicity system, Millipore Co.); tris(hydroxymethylamino)methane (TRIS) p.a. (Acros, Belgium); NaCl p.a. (Sigma-Aldrich, USA); NaN₃ p.a. (Acros, Belgium); ammonium acetate p.a. (Acros, Belgium).

Results and Discussion

The present copolymers were prepared by copolymerization of methacrylic acid (MAA) and a poly(ethylene oxide) macromonomer, see Scheme 1. The macromonomer (MPEG-MM) was synthesized from poly(ethylene glycol) monomethyl ether (MPEG) and MAA. Accordingly, in addition to the macromonomer the reaction product may contain residual MPEG and MAA. The copolymerization of MAA and the macromonomer was initiated with the free-radical initiator sodium peroxodisulfate. The resulting copolymers are distributed in at least three parameters: (1) the total molar mass, (2) the copolymer composition, and (3) the chain length of the PEO grafts. In addition, the copolymer can be contaminated by small amounts of homopolymers poly(methacrylic acid) (PMAA) and P(MPEG-MM).

For a detailed analysis of the course of the reaction, the different starting and intermediate materials have been analyzed by liquid chromatography. As the macromonomer and the precursor MPEG used in the present study are technical products, it was important to analyze their purity.



Scheme 1. Copolymerization of methacrylic acid and poly(ethylene glycol) macromonomer.

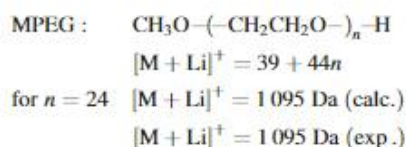
Such analysis can be conducted by liquid chromatography at critical conditions (LC-CC). At the critical point of adsorption for PEG, LC-CC separates MPEG and the macromonomer with regard to their endgroups irrespective of molar mass. Typical LC-CC conditions for PEO are a non-polar stationary phase of RP-18 (octadecyl modified silica gel) and a mobile phase composition of methanol-water 81:19 wt.-%.

At critical conditions, PEGs of different molar masses elute at the same elution volume. This is indicated in Figure 1A for a molar mass range of 400–35 000 g/mol. The starting PEG and the resulting MPEG elute under these chromatographic conditions in single uniform peaks, see Figure 1B and C. Complemented by MALDI-TOF analyses, this clearly indicates that both samples are homogeneous with regard to their endgroups and only polydisperse regarding molar mass.

The analysis of two different reaction products of MPEG and MAA is presented in Figure 2 and compared to the elution pattern of MPEG. The chromatograms of MPEG-MM 1 and 2 indicate directly that the chemical compositions of the two reaction products are different. While MPEG-MM 2, exhibits one major elution peak at 2.8 mL with a very small second peak at 2.5 mL, MPEG-MM 1 has a trimodal distribution with a small peak eluting at 1.8 mL, followed by peaks at 2.5 mL and 2.8 mL. In this case the peak at 2.5 mL is quite significant.

A comparison of the peak elution volumes in Figure 2A–C indicates that the elution peak at 2.5 mL could correspond to MPEG. To verify this assumption, the samples were subjected to MALDI-TOF mass spectrometry for analysis. The measurements were carried out in the positive-linear mode with dithranol as the matrix. As cation donor LiCl was added and, accordingly, the formation of $[M+Li]^+$ molecular ions can be assumed.

The MALDI-TOF spectrum of MPEG in Figure 3A shows one uniform oligomer distribution with the highest peak intensity ranging from 1000–1150 Da which is in good agreement with the nominal molar mass of about 1000 g/mol. The mass peaks can be attributed to poly(ethylene glycol) monomethylether oligomers:



The fact that no other oligomer series is detected complements the LC-CC result that the sample is uniform with regard to the endgroups. A similarly uniform MALDI-TOF spectrum is obtained for the macromonomer MPEG-MM 2, see Figure 3C. The oligomer series in this case can be assigned to the expected macromonomer structure and is in perfect agreement with the LC-CC result:

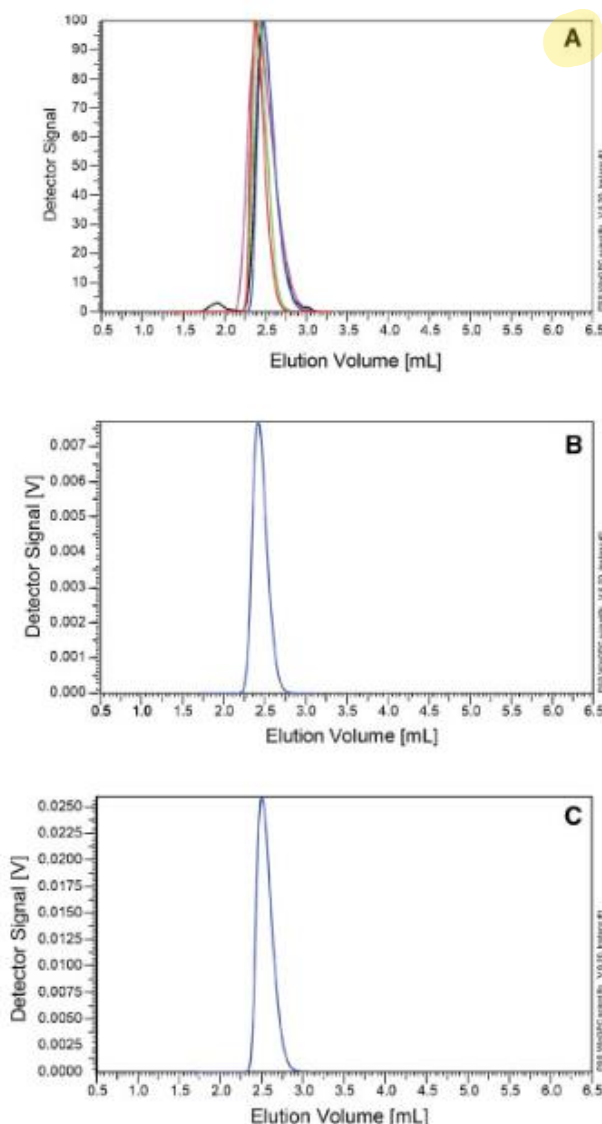
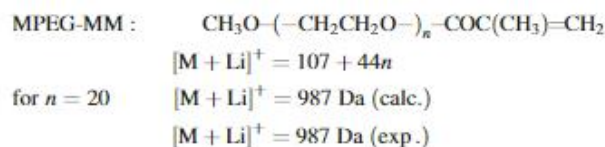


Figure 1. LC-CC elugrams of PEGs of different molar masses (red: 400, green: 970, blue: 6000, black: 12 000, pink: 35 000 g/mol) (A), PEG 970 (B) and MPEG (C), stationary phase: Knauer Nucleosil RP18-100, mobile phase: MeOH/water 81:19 wt.-%, detector: ELSD.



For macromonomer MPEG-MM 1 a more complex MALDI-TOF spectrum is obtained, see Figure 3B. The major oligomer series indicated by (X) can, similar to MPEG-MM 2, be attributed to the macromonomer oligomers. The second oligomer series (■) that appears in the

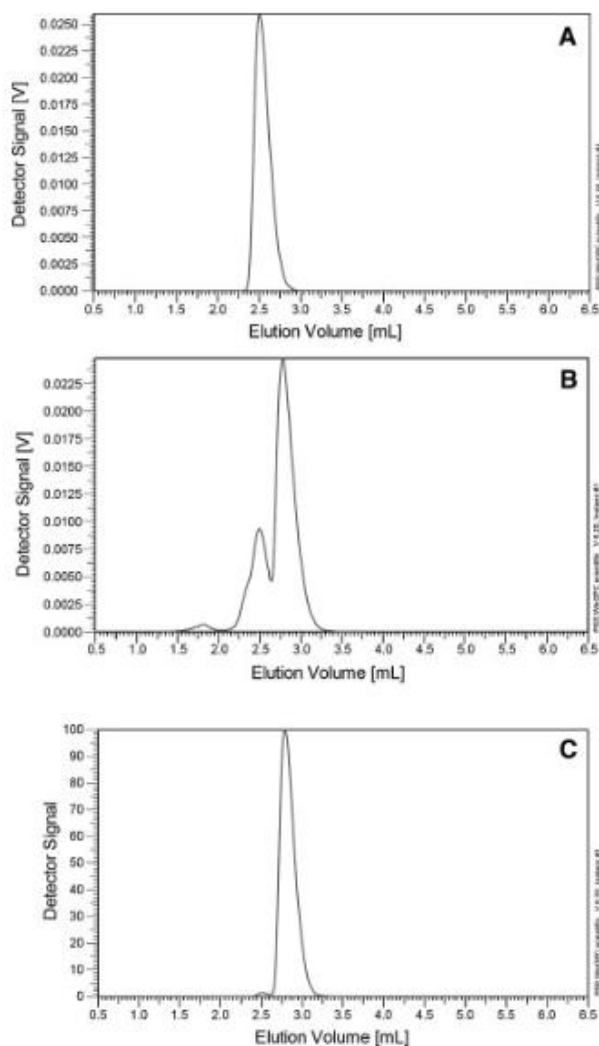


Figure 2. LC-CC elugrams of MPEG (A), MPEG-MM 1 (B) and MPEG-MM 2 (C), chromatographic conditions see Figure 1.

spectrum at a rather low intensity can be assigned to residual MPEG as is indicated in the corresponding chromatogram, see Figure 2B. The chromatographic fraction appearing at 1.8 mL in Figure 2B was too small to be detected by MALDI-TOF mass spectrometry.

The amounts of by-products in the samples MPEG-MM can be estimated roughly from the peak areas of the elution peaks in Figure 2. The concentration of the main by-product MPEG is roughly 29 wt.-% for MPEG-MM 1 and 2 wt.-% for MPEG-MM 2, while the concentration of the by-product at 1.8 mL is less than 2 wt.-% in MPEG-MM 1 and not detectable in MPEG-MM 2.

To investigate the polymerization behavior of the macro-monomer, sample MPEG-MM 2 was polymerized to form the homopolymer poly(MPEG-MM 2) (sample 1). The molar mass distribution of this homopolymer was analyzed

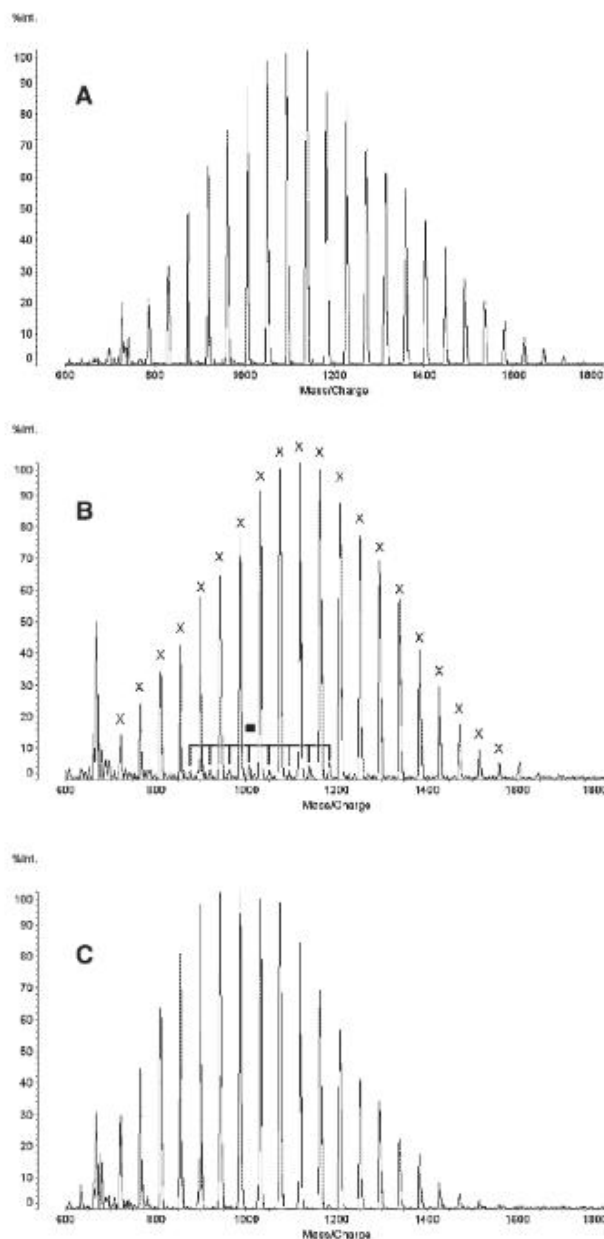


Figure 3. MALDI-TOF spectra of MPEG (A), MPEG-MM 1 (B) and MPEG-MM 2 (C).

by aqueous SEC on a stationary phase of PSS Suprema linear M (300 × 8 mm i.d.) with an aqueous eluent of the following composition: water, TRIS (tris(hydroxymethyl)aminoethane) 0.08 mol/L, NaCl 0.15 mol/L, Na₂S₂O₃ 0.01 mol/L. The flow rate was 1 mL/min and the detection was carried out with a RI detector. The SEC elution profile of the sample is shown in Figure 4.

The elution pattern indicates a separation of different oligomers. The peak at 12.4 mL corresponds to the system peak. This peak overlaps with the monomer peak. The

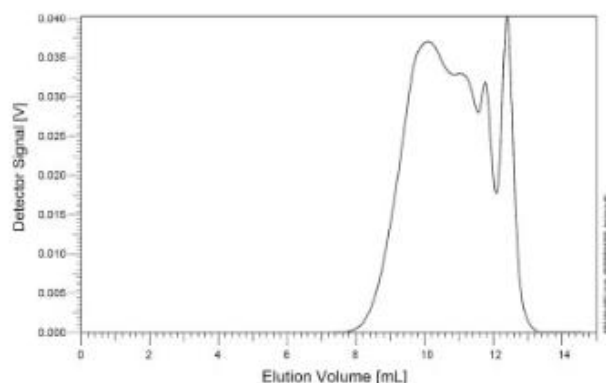


Figure 4. SEC elugram of sample 1, stationary phase: Suprema linear M, mobile phase: water, TRIS 0.08 mol/L, NaCl 0.15 mol/L, NaN_3 0.01 mol/L, detector: RI.

lowest observable oligomer at 11.8 mL is due to the dimer, the shoulder at 11.2 mL contains oligomers from the trimer to the hexamer. At elution volumes less than 10.5 mL higher oligomers are eluted.

For a more specific analysis of the homopolymer a LC-CC separation was conducted using the same chromatographic conditions that have been used for MPEG and MPEG-MM. The corresponding chromatogram in Figure 5 indicates that the sample is not homogeneous but contains species of different chemical compositions.

The latest eluting small peak appears at roughly 2.5 mL and could correspond to residual MPEG that was found as a by-product in the macromonomer, see Figure 2. Another small peak elutes at about 2.1 mL while the major part of the sample elutes at 1.0–2.0 mL. To analyze the chemical composition of the different elution peaks the sample was fractionated and the fractions analyzed by MALDI-TOF mass spectrometry. The significant advantage of MALDI-TOF over other spectroscopic techniques is that very frequently only an analytical LC-CC separation is required to accumulate sufficient amounts for analysis.

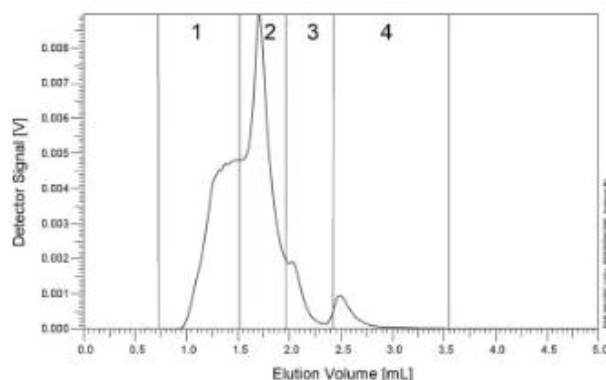


Figure 5. LC-CC elugram of sample 1, chromatographic conditions see Figure 1, fractionation into four fractions is indicated.

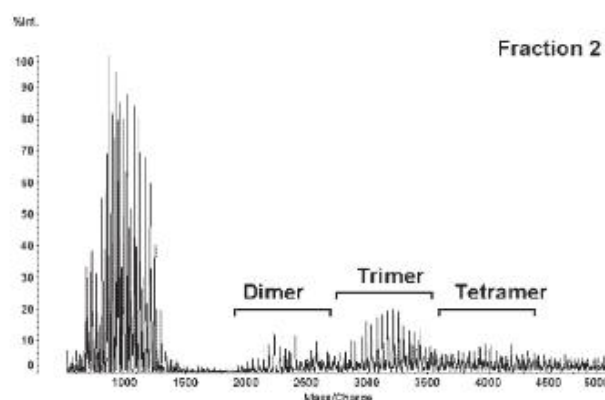
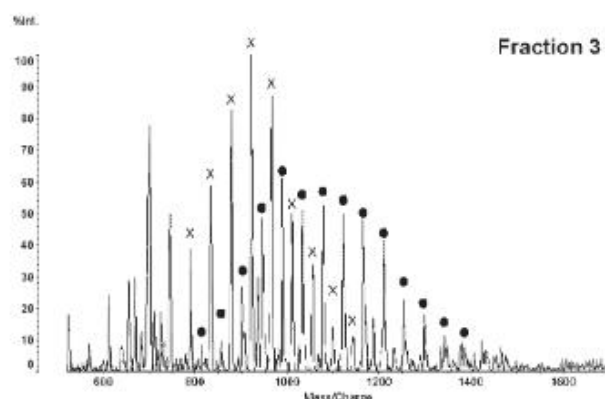
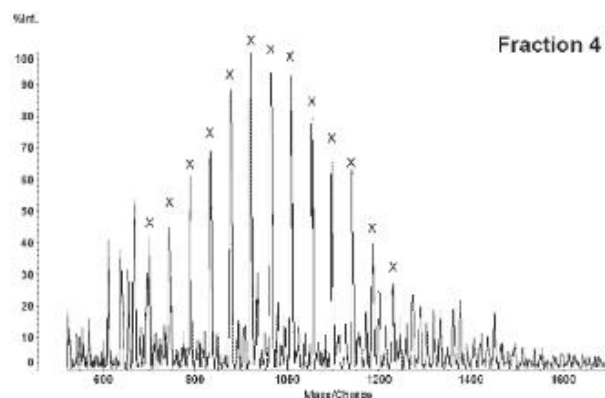


Figure 6. MALDI-TOF spectra of fractions 2–4 of sample 1 from LC-CC separation in Figure 5.

As was expected, fraction 4 is due to residual MPEG. This compound is also found to a certain extent in fraction 3, see oligomer series (x) in Figure 6. In addition to MPEG the spectrum of fraction 3 indicates a second (major) oligomer series that corresponds to non-reacted macromonomer MPEG-MM (●). The most interesting spectrum is obtained for fraction 2, see Figure 6. In addition to oligomer series for MPEG and MPEG-MM in the low mass range of 600–1300 Da, three new oligomer series A, B, C are obtained at higher masses from 1900 Da on. The peak-to-peak mass increments are 44 Da and indicate MPEG-MM derivatives.

The mass differences between the oligomer series are roughly 1 000 Da and correspond to the molar mass of the macromonomer. It can, therefore, be assumed that the oligomer series A, B, and C belong to multiples of MPEG-MM, i.e., the dimer (A), trimer (B), and tetramer (C), respectively. It can be assumed that fraction 1 contains higher molar mass oligomers. These could not be detected by MALDI-TOF, however, due to low concentration, high molar mass, and high polydispersity.

As has been shown, LC-CC can be used very efficiently to detect by-products in the present macromonomer and homopolymer. In the next step, the copolymerization of MPEG-MM with methacrylic acid shall be analyzed. The copolymerization was conducted in an aqueous solution with sodium persulfate as initiator. The feed composition was varied between 10, 20, and 30 wt.-% of methacrylic acid (samples 2, 3, and 4). Such composition corresponds to molar compositions of 58, 72, and 84 mol-% methacrylic acid in the feed.

The separation of the resulting copolymers by LC-CC is shown in Figure 7. Again, critical conditions for PEG are used to compensate for the effect of the PEO molar mass distribution.

As could be expected, the copolymers yield chromatograms that are very similar to the chromatogram of Poly(MPEG-MM 2). A small elution peak at 2.5 mL shows that the samples contain residual MPEG. Due to the higher polarity of the copolymer fractions, these elute before the MPEG at elution volumes of 1.0–2.2 mL. From the peak areas and a corresponding calibration, the amounts of residual MPEG can be determined quantitatively. Accordingly, sample 1 contains 3 wt.-% of MPEG, while samples 2 and 3 contain both 2 wt.-% of MPEG.

The copolymer fractions eluting before MPEG at elution volumes of 1.0–2.2 mL exhibit trimodal distributions that can be caused by multimodalities in molar mass or in chemical composition. To detect differences in chemical composition, the LC-CC separation is combined with FTIR detection through the LC-Transform interface. The design concept and the operation of the interface is described in refs.^[21–23] Briefly, the eluate leaving the chromatographic separation system is sprayed onto a rotating Germanium disc through a heated nebulizer nozzle. The mobile phase evaporates and the polymer fractions are deposited as tracks of solid material. After depositing all LC-CC fractions, the Germanium disc is placed into the transmission part of the FTIR spectrometer and spectra are taken from all positions of the disc. As the result of the measurements, spectra of all fractions are obtained that are used for identification of the chemical structures. Concentration profiles of different chemical species can be obtained by plotting specific FTIR intensities as a function of elution volume. The total concentration profile (Gram-Schmidt plot) is obtained by summarizing all intensities over all wavenumbers. The concentration profile of the ethylene glycol units is meas-

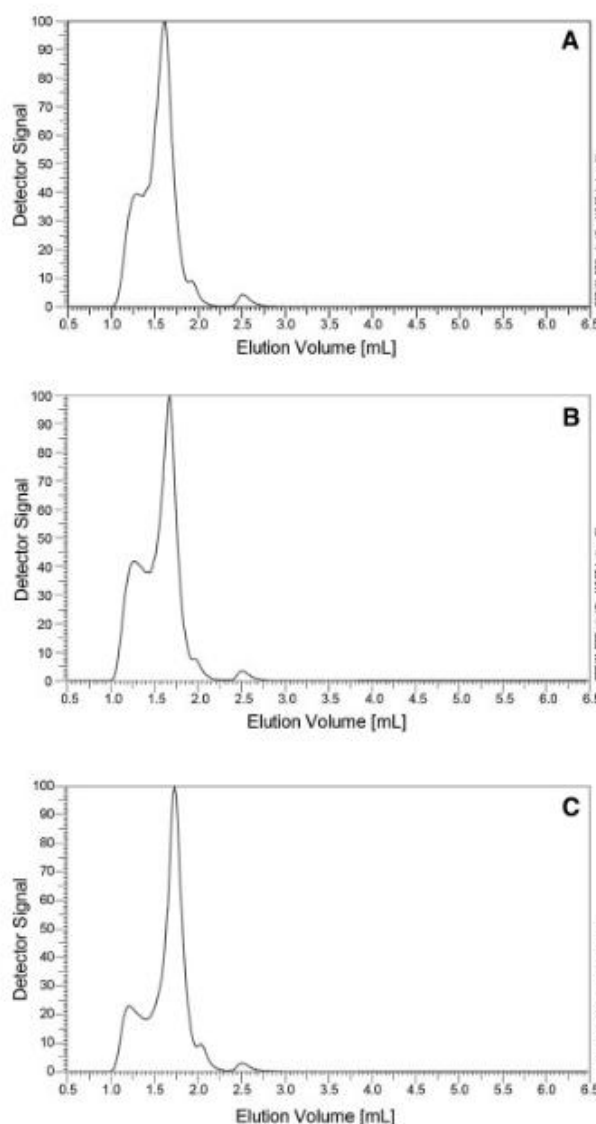


Figure 7. LC-CC elugrams of copolymers of MPEG-MM and MAA, chromatographic conditions see Figure 1, samples: 2 (A), 3 (B), 4 (C).

ured at a wavenumber of $1\,100\text{ cm}^{-1}$, while the concentration profile of the methacrylic acid units is measured at a wavenumber of $1\,580\text{ cm}^{-1}$.

The results of the LC-FTIR analysis of sample 4 is shown in Figure 8. As can be seen, the concentration profiles for EG and MAA are very similar indicating a rather homogeneous distribution of the two monomers across the elution peak. A detailed analysis of the concentration profiles indicates that the earlier eluting fractions contain higher amounts of MAA.

By far the deepest insight into the complexity of the present copolymers can be obtained by 2D chromatography. On-line coupled 2D chromatography combines the

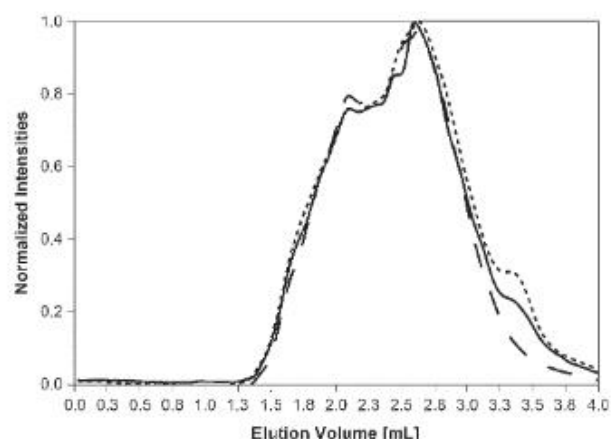


Figure 8. LC-CC/FTIR analysis of sample 4, chromatographic conditions see Figure 1, Gram-Schmidt plot (solid), EG (dotted), MAA (dashed).

separation with regard to chemical composition in the first dimension with the analysis of the resulting fractions regarding molar mass in the second dimension. In the present case, LC-CC is used for separating the copolymer samples into the true copolymer and residual precursors or by-products such as MPEG or the macromonomer.

2D chromatography requires a powerful detector for the second dimension. For applications with organic solvents like tetrahydrofuran an evaporative light scattering detector (ELSD) can easily be used. This is not possible for aqueous SEC at the previously discussed experimental conditions. Due to the salt-containing eluent the spray nozzle of the detector gets blocked. A RI detector cannot be used in 2D-LC because the system uses different eluents in both dimensions. The detection problem was solved by using ammonium acetate as a volatile salt. Using water-ammonium acetate as the mobile phase it became possible to use the ELSD as detector.^[24]

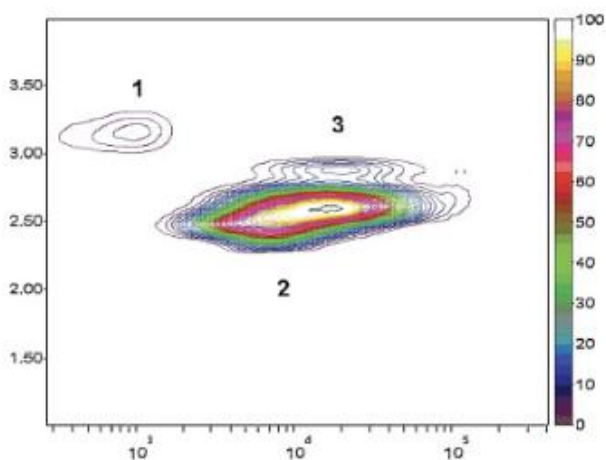


Figure 9. 2D separation of sample 1, 1st dimension: LC-CC, 2nd dimension: SEC.

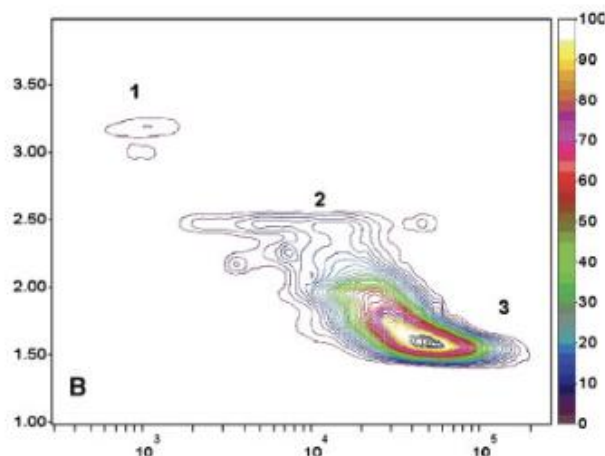
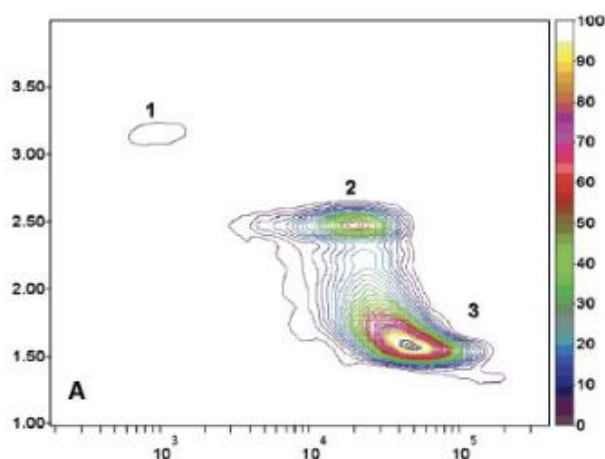


Figure 10. 2D separations of samples 3 (A) and 4 (B), 1st dimension: LC-CC, 2nd dimension: SEC.

The 2D analysis of sample 1 is shown in Figure 9. In vertical direction the sample is separated into chemically different fractions by LC-CC while the horizontal direction represents the molar mass separation. Obviously, the sample contains three chemically different species with two of them being similar in molar mass.

Species 1 with a molar mass of roughly 1 000 g/mol can be assigned readily to the precursor MPEG that was already detected by LC-CC in Figure 5 and MALDI-TOF analysis. Species 2 as the major component of the sample can be assigned to the true poly(MPEG-MM 2) homopolymer. The molar mass determined from the 2D experiment is $\bar{M}_w = 17\,000$ g/mol and in good agreement with the SEC analysis given in Figure 4. Finally, species 3 has a molar mass close to the homopolymer ($\bar{M}_w = 12\,500$ g/mol) but is shifted towards higher elution volumes in the first dimension indicating a lower polarity. The chemical structure of this fraction is subject to further investigations. The relative concentrations of the different species are 1.5% (1), 95.8% (2), and 2.1% (3), respectively. This clearly indicates that sample 1 is rather pure homopolymer.

Table 2. Composition of samples 3 and 4 as determined by 2D-LC.

Sample	MPEG		Homopolymer		Copolymer	
	Amount	$\bar{M}_w$	Amount	$\bar{M}_w$	Amount	$\bar{M}_w$
	%	g/mol	%	g/mol	%	g/mol
3	0.4	980	19.6	16 500	79.4	40 900
4	0.7	1 000	5.3	8 600	93.6	40 100

Similar to the homopolymer, copolymers of different compositions were analyzed by 2D chromatography, see Figure 10. The appearance of the resulting contour plots is significantly different from the contour plot of the homopolymer. As has been experienced in a number of other cases, the resolution of the first dimension (LC-CC) is significantly improved. This is, apparently, due to the fact that in the 2D experiments the flow rate of the first dimension is significantly lower as compared to standard LC-CC experiments.

The 2D contour plot for sample 3 is shown in Figure 10A. Similar to the homopolymer in Figure 9, this sample also contains minor amounts of MPEG that appear in the contour plot as species 1. The molar mass of this fraction is roughly 1 000 g/mol. In the high molar mass region two species are obtained, species 2 having a molar mass of ca. 16 000 g/mol and species 3 with a molar mass of ca. 40 000 g/mol. The chromatographic behaviour of species 2 is very similar to the behaviour of poly(MPEG-MM 2) as can be seen in Figure 9. It is, therefore, tentatively assumed that the sample contains a fraction of homopolymerized macromonomer. Accordingly, species 3 can be assigned to the copolymer fraction. As is expected, the molar mass is higher as compared to the homopolymer fraction. It is interesting to mention that the projection of the copolymer peak has a positive slope towards lower molar masses. This indicates that separation in the LC-CC dimension takes place according to decreasing molar mass. This finding is in perfect agreement with the theory of LC-CC that predicts an SEC mode for the polymer backbone at critical conditions of the PEO grafts.^[6]

The contour plot for sample 4 in Figure 10B indicates that this sample contains only minor amounts of homopolymer. This is in agreement with the higher concentration of MAA in the monomer feed that results in the preferential formation of copolymer molecules.

The quantitative composition of the two samples is summarized in Table 2.

Conclusion

It has been shown that hydrophilic copolymers can be analyzed by different methods of liquid chromatography. LC-CC yields information on the chemical composition while 2D-chromatography provides information on molar mass as a function of chemical composition. It is shown

that the copolymerization of methacrylic acid and a PEG-macromonomer (PEG-MM) results in the formation poly(PEG-MM) in addition to the copolymer.

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